# PhET Activity: Work and Energy with the Spring

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## 1 Introduction

Our task is to predict the work required to compress a spring. This calculation is relevant for a multitude of physical systems that *oscillate.* The work done by compressing a spring by $\Delta\vec{x}$ is

$$\Delta W = \vec{F} \cdot \Delta\vec{x} \tag{1}$$

The force required by Newton's 3rd law is $\vec{F} = k\Delta\vec{x}$, the opposite of the spring force. Combining equations, we find

$$\Delta W = k\Delta\vec{x} \cdot \Delta\vec{x} = k(\Delta x)^2 \tag{2}$$

**There is one complication:** each time we displace the spring another $\Delta x$, the force changes, so $\Delta W$ changes. We must *add up* all the contributions $\Delta W$ to obtain the total work done, $W$.

## 2 Simulation of Spring Behavior

Load the following PhET simulation: `https://phet.colorado.edu/en/simulations/hookes-law`. Click on the energy tab. Click on the Force Plot button, and add the energy by checking the Energy box. We see a linear plot appear that displays the force on the y-axis and the displacement on the x-axis.

Set the spring constant to 300 N m$^{-1}$, and the displacement to zero. Fill in Tab. 1 below with each $\Delta x$, the *change* in position, $\Delta W$, the *change* in total energy, and the cumulative value for $W$, assuming $k = 300$ N m$^{-1}$. The cumulative value $W$ is a sum of all previous $\Delta W$ values.

| $\Delta x$ (cm) | $\Delta W$ (J) | $W$ (J) |
|---|---|---|
| 0.1 | | |
| 0.2 | | |
| 0.3 | | |
| 0.4 | | |
| 0.5 | | |
| 0.6 | | |
| 0.7 | | |
| 0.8 | | |
| 0.9 | | |
| 1.0 | | |

Table 1: (Left) Change in position. (Center) Change in work done. (Right) Total work done.

## 3 Analysis of Results

**Create a graph of $W$ versus** $x$, that is, a graph of spring position versus cumulative work done. Now add a quadratic curve of the following form to your graph: $W = \frac{1}{2}kx^2$. Note that the *area* under the curve is the work done, because for each little change $\Delta x$, we add $\vec{F} \cdot \Delta\vec{x}$ to the total work done. Can you see why the formula $W = \frac{1}{2}kx^2$ works geometrically, from the graph of Applied Force versus position in the simulation?